

PE-EC703C — Wavelet Transforms

Semester VII • 3L:0T:0P • 3 Credits

Module 2 Exam Notes: Continuous Wavelet Transform (CWT)

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1. Continuous Time Wavelets & Definition of CWT

The Continuous Wavelet Transform (CWT) maps a 1D time-domain signal into a 2D time-scale plane, allowing localized spectral analysis.

Definition of CWT

The **Continuous Wavelet Transform (CWT)** of a continuous-time signal $x(t) \in L^2(\mathbb{R})$ is the correlation of the signal with a scaled and translated version of a mother wavelet function $\psi(t)$:

$$CWT_x(a, b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} x(t) \psi^* \left(\frac{t-b}{a} \right) dt$$

where:

- $a \in \mathbb{R} \setminus \{0\}$ is the **scale parameter** (controls compression/dilation).
- $b \in \mathbb{R}$ is the **translation parameter** (controls shift along the time axis).
- $\psi(t)$ is the **mother wavelet**, and $\psi^*(t)$ denotes its complex conjugate.
- $\frac{1}{\sqrt{|a|}}$ is the normalization factor ensuring energy is invariant across scales.

2. Scaling and Translation Parameters

- **Scaling (a):** Scaling either dilates (stretches) or compresses the mother wavelet:
 - **Large scale** ($|a| > 1$): Wavelet is dilated, representing a low-frequency, long-duration window (captures coarse features).
 - **Small scale** ($|a| < 1$): Wavelet is compressed, representing a high-frequency, short-duration window (captures fine details and transients).
- **Translation (b):** Moves the wavelet along the time axis. It indicates the time center of the window.

3. Constant Q-Factor Filtering Interpretation & Time-Frequency Resolution

The ratio of the center frequency f_c of a filter to its bandwidth Δf is defined as the **Quality Factor (Q-factor)**:

$$Q = \frac{f_c}{\Delta f}$$

In CWT, compressing the wavelet in time by a stretches its Fourier spectrum by $1/a$. Thus, center frequency and bandwidth scale proportionally:

$$f_c(a) = \frac{f_c(1)}{a}, \quad \Delta f(a) = \frac{\Delta f(1)}{a} \implies Q = \frac{f_c(a)}{\Delta f(a)} = \text{Constant}$$

Hence, the CWT behaves as a **constant-Q bandpass filter bank**.

Wavelet Dilation and Compression

Mother Wavelet ($a = 1, b = 0$) **Dilated Wavelet** ($a = 2.0, b = 0$) **Compressed & Shifted** ($a = 0.5, b = 3.0$)



4. Mathematical Properties of CWT

4.1 CWT as a Linear Operator

The CWT is a **linear operator**. If $x(t) = c_1x_1(t) + c_2x_2(t)$, then:

$$CWT_x(a, b) = c_1CWT_{x_1}(a, b) + c_2CWT_{x_2}(a, b)$$

4.2 Admissibility Condition

For a function $\psi(t)$ to be a valid mother wavelet, it must satisfy the **admissibility condition**:

$$C_\psi = \int_{-\infty}^{\infty} \frac{|\Psi(f)|^2}{|f|} df < \infty$$

where $\Psi(f)$ is the Fourier transform of $\psi(t)$.

- **Consequence:** For the integral to be finite, $\Psi(0) = 0$. This implies:

$$\int_{-\infty}^{\infty} \psi(t) dt = 0$$

The average value (or DC component) of a wavelet must be **zero**. It must oscillate above and below the zero axis.

4.3 Inverse Continuous Wavelet Transform

If the admissibility condition $C_\psi < \infty$ is met, the signal $x(t)$ can be reconstructed from its wavelet coefficients $CWT_x(a, b)$ via the inverse CWT formula:

$$x(t) = \frac{1}{C_\psi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} CWT_x(a, b) \frac{1}{\sqrt{|a|}} \psi\left(\frac{t-b}{a}\right) \frac{dadb}{a^2}$$

4.4 Energy Conservation (Parseval's Relation)

The wavelet transform preserves energy. If $x(t) \in L^2(\mathbb{R})$, then:

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{C_\psi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |CWT_x(a, b)|^2 \frac{dadb}{a^2}$$

Quick Revision — Module 2

Continuous Wavelet Transform (CWT) | PE-EC703C

Key Wavelet Equations

Continuous Wavelet Transform	$CWT_x(a, b) = \frac{1}{\sqrt{ a }} \int_{-\infty}^{\infty} x(t) \psi^* \left(\frac{t-b}{a} \right) dt$
Admissibility Condition	$C_\psi = \int_{-\infty}^{\infty} \frac{ \Psi(f) ^2}{ f } df < \infty \implies \int_{-\infty}^{\infty} \psi(t) dt = 0$
Inverse Wavelet Transform	$x(t) = \frac{1}{C_\psi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} CWT_x(a, b) \frac{1}{\sqrt{ a }} \psi \left(\frac{t-b}{a} \right) \frac{da db}{a^2}$
Constant Q-Factor	$Q = \frac{f_c}{\Delta f} = \text{Constant} \quad (\text{Proportional frequency/bandwidth scaling})$

One-Line Definitions

- **CWT:** The correlation of a continuous-time signal with a continuously scaled and translated mother wavelet.
- **Scale Parameter (a):** Controls the compression (small a , high frequency) or dilation (large a , low frequency) of a wavelet.
- **Translation Parameter (b):** Determines the temporal location of the wavelet window center along the time axis.
- **Constant Q-Factor:** Quality factor $Q = f_c / \Delta f$ remains constant across scales, behaving as a logarithmic bandpass filter bank.
- **Admissibility Condition:** A mathematical constraint requiring a wavelet's DC component to be zero for reconstruction to be possible.
- **Mother Wavelet:** The prototype wavelet function used to generate all shifted and scaled daughter wavelets.

Frequently Asked / Viva Questions

Q1. State the physical meaning of the scale parameter ' a '.

Scale ' a ' is inversely proportional to frequency. A small scale ($a < 1$) compresses the wavelet, corresponding to high-frequency transients. A large scale ($a > 1$) dilates the wavelet, corresponding to slow-varying low-frequency components.

Q2. What is the significance of the admissibility condition?

The admissibility condition ensures that the wavelet does not carry a DC component ($\int_{-\infty}^{\infty} \psi(t) dt = 0$). This guarantees that the inverse wavelet transform exists and enables perfect reconstruction of the signal.

Q3. Explain the constant Q-factor filtering property of the CWT.

In CWT, as scale a changes, the center frequency and bandwidth of the wavelet

Frequently Asked / Viva Questions (contd.)

filter change proportionally. Since $Q = f_c/\Delta f$, the quality factor remains constant across all scales, behaving as a logarithmic bandpass filter bank.

Q4. What is the difference between Gabor transform and Wavelet transform?

The Gabor transform is a special case of STFT using a Gaussian window. It uses a fixed window envelope across all frequencies. The Wavelet transform uses a scaled envelope that compresses at higher frequencies and expands at lower frequencies.